A PROPERTY SMOOTH LOCAL ON SOURCE AND TARGET, STABLE UNDER POSTCOMPOSITION WITH SMOOTH MORPHISMS, IS SMOOTH LOCAL ON THE SOURCE-AND-TARGET

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two lemmas and one definition of the same chapter, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Smooth local on the source: precomposition stability. Stacks Project, Descent on Spaces, Lemma lemma-precompose-property-local-source. Supplies that P is stable under precomposing with smooth morphisms.
- Smooth local on the target: base change stability. Stacks Project, Descent on Spaces, Lemma lemma-pullback-property-local-target. Supplies that P is stable under smooth base change.
- Smooth local on the source-and-target: definition. Stacks Project, Descent on Spaces, Definition definition-local-source-target. Supplies part (3)(b) being assumed and part (3) being proved.

Source reference. Original lemma 06EQ: sources/stacks-chow/tags.txt, source lines 8224–8224. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 06EM: sources/stacks-chow/tags.txt, source lines 8221–8221. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 06F8: sources/stacks-chow/tags.txt, source lines 8242–8242. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces over S. Assume*

(1) *$\mathcal{P}$ is smooth local on the source,*
(2) *$\mathcal{P}$ is smooth local on the target, and*
(3) *$\mathcal{P}$ is stable under postcomposing with smooth morphisms: if $f : X \to Y$ has $\mathcal{P}$ and $Y \to Z$ is a smooth morphism then $X \to Z$ has $\mathcal{P}$.*

Then $\mathcal{P}$ is smooth local on the source-and-target.

Proof. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces which satisfies conditions (1), (2) and (3) of the lemma. By Lemma 06EQ we see that $\mathcal{P}$ is stable under precomposing with smooth morphisms. By Lemma 06EM we see that $\mathcal{P}$ is stable

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under smooth base change. Hence it suffices to prove part (3) of Definition 06F8 holds.

More precisely, suppose that $f : X \to Y$ is a morphism of algebraic spaces over S which satisfies Definition 06F8 part (3)(b). In other words, for every $x \in X$ there exists a smooth morphism $a_x : U_x \to X$, a point $u_x \in |U_x|$ mapping to x, a smooth morphism $b_x : V_x \to Y$, and a morphism $h_x : U_x \to V_x$ such that $f \circ a_x = b_x \circ h_x$ and h_x has $\mathcal{P}$. The proof of the lemma is complete once we show that f has $\mathcal{P}$. Set $U = \coprod U_x$, $a = \coprod a_x$, $V = \coprod V_x$, $b = \coprod b_x$, and $h = \coprod h_x$. We obtain a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow[h]{} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with a, b smooth, a surjective. Note that h has $\mathcal{P}$ as each h_x does and $\mathcal{P}$ is smooth local on the target. Because a is surjective and $\mathcal{P}$ is smooth local on the source, it suffices to prove that $b \circ h$ has $\mathcal{P}$. This follows as we assumed that $\mathcal{P}$ is stable under postcomposing with a smooth morphism and as b is smooth. $\square$